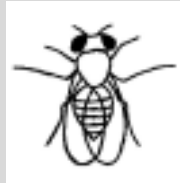
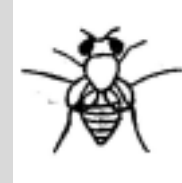
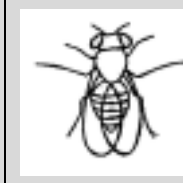


7(a)	<i>allele</i> – variety / form, of a gene ; <i>dominant</i> –(allele) always expressed / always has effect on the phenotype / expressed in heterozygote / has effect on the phenotype of heterozygote ;	2
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Question	Answer					Marks	
7(b)(i)		wild type parent		double homozygous recessive parent		5	
	parental phenotype	red eye, long wing		purple eye, vestigial wing			;
	parental genotype	RrNn		rrnn			;
	offspring						
	offspring phenotype	red eye, long wing	purple eye, vestigial wing	red eye, vestigial wing	purple eye, long wing		;
	offspring genotype	RrNn	rrnn	Rrnn	rrNn		::
	numbers of offspring	1339	1195	151	154		
	offspring genotypes – 4 correct = 2 marks, 3 correct = 1 mark allow short wing for vestigial wing allow alleles in any order in genotypes ignore brackets and circles around alleles						

Question	Answer	Marks
7(b)(ii)	any three from: 1 (two) genes are linked / autosomal linkage or <u>alleles</u> inherited together ; 2 <u>no</u>, independent / random, assortment ; 3 large numbers of parental type offspring / small numbers of recombinant offspring ; 4 recombinants due to crossing over ; 5 during, meiosis / prophase 1 / gamete formation ;	3

Question	Answer	Marks
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